

Plant Disease classification using Deep Learning

Submitted in partial fulfillment of the requirements for the degree of

Master of Science In Data Science

by

SUDDALA RAMESH

Reg no: 24MDT0193

Under the guidance of

Prof. Venkateswarlu B

Department of Mathematics

School of Advanced Sciences

VIT - Vellore



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November, 2025

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I hereby declare that the thesis entitled "**Plant Disease classification using Deep Learning**" submitted by me, for the award of the degree of *Master of Science in Data Science* to VIT is a record of Bonafide work carried out by me under the supervision of **Prof. VENKATESWARLU B.**

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Suddala Ramesh

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Place : Vellore

Date :

Signature of the Guide

Prof. VENKATESWARLU B

Department of Mathematics

SAS, VIT- Vellore

Name & Signature of the Examiner

Dr. KHADARBABU S K

Head, Department of Mathematics

SAS, VIT-Vellore

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Signature of the Student

SUDDALA RAMESH

ABSTRACT

Plant diseases have a substantial impact on global agricultural productivity, often resulting in economic losses and diminished crop quality. Early identification of plant leaf diseases is critical for timely intervention and prevention. Traditional methods rely on manual inspection, which is slow, costly, and susceptible to human error. Recent advancements in deep learning provide highly effective solutions for automated plant disease detection utilizing image classification. This project presents a deep learning-based plant disease classification system utilizing the PlantVillage dataset comprising 38 classes of diseased and healthy leaves. A transfer learning approach employing the MobileNetV2 architecture was employed to achieve robust performance with limited computational resources. The model underwent training in two phases: an initial frozen feature extraction phase followed by fine-tuning of deeper layers to enhance classification accuracy. A Streamlit web application was developed to provide an interactive and user-friendly interface for real-time disease prediction. Users can upload leaf images or capture images using a webcam. The system preprocesses the input, performs inference, and displays the predicted disease class along with confidence scores and detailed disease descriptions, including recommended treatments. The application demonstrates consistent performance across several plant species such as tomato, potato, apple, grape, and citrus. This work demonstrates the effectiveness of lightweight deep learning models in agricultural disease detection and highlights the potential for scalable deployment in precision farming, mobile advisory systems, and real-world agricultural support platforms.

Keywords: *plant disease classification , streamlit, fine tuning, diagnosis, crops, treatment, diseased leafs, Image processing, feature extraction, transfor learning.*

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List of Abbreviations

Abbreviations	Full form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Networks
ReLU	Rectified Linear Unit
TPU	Tensor Processing Unit
GUI	Graphical User Interface
TL	Transfer Learning
TF	Tensor Flow
TFLite	TensorFlow Lite
API	Application Programming Interface
HDF5	Hierarchical Data Format Version 5
IoT	Internet Of Things

1. INTRODUCTION

1.1 Overview of Plant Disease Classification

Agriculture serves as the foundation of global food security, and plant health is crucial for maintaining crop productivity. Plant diseases, if not identified promptly, can spread rapidly and result in significant losses in yield and quality. Conventional disease identification methods rely on expert inspection, which is often time-consuming, subjective, and inaccessible to farmers in remote regions.

The advent of machine learning and deep learning has made automated disease detection using leaf images a practical and highly accurate solution. Convolutional Neural Networks (CNNs), particularly models based on transfer learning such as MobileNetV2, have demonstrated effectiveness in classifying plant diseases with minimal computational resources. These systems facilitate faster diagnosis, support large-scale agricultural monitoring, and enhance decision-making for disease management.

This project aims to develop a robust plant disease classification model utilizing deep learning, trained on the PlantVillage dataset. The project also involves creating a Streamlit-based web application that enables real-time disease prediction through image upload or webcam input. In addition to predictions, the system provides confidence scores and treatment recommendations to support practical disease management.

1.2 Problem Statement

Farmers frequently encounter challenges in accurately and timely identifying plant diseases to mitigate potential damage. Inaccurate or delayed diagnoses can result in inappropriate treatment, diminished crop yields, and financial losses. Manual identification necessitates specialized expertise, which may not always be readily available.

Consequently, there is a pressing need for an automated, precise, and user-friendly system capable of identifying plant diseases from leaf images and providing users with pertinent information for corrective measures.

1.3 Need for Intelligent Disease Detection

The significance of automated plant disease detection is substantiated by several practical challenges:

- Limited availability of agricultural experts in rural areas

Many farming communities lack access to plant pathologists or extension officers.

- Time-sensitive nature of disease progression

Delayed response exacerbates the spread and severity of infections.

- High cost of laboratory tests

Traditional testing necessitates equipment and resources that may not be financially feasible

- Variability in human diagnosis

Subjective judgment can result in inconsistent outcomes.

An AI-powered system offers consistent, scalable, and expedient disease identification, rendering it practical for routine agricultural applications.

1.4 Objectives of the Study

The major objectives of this project are:

1. Design and train a deep learning model capable of accurately classifying plant diseases using image data.
2. Utilize the PlantVillage dataset, which comprises 38 plant disease and healthy classes.
3. Implement transfer learning with MobileNetV2 and explore fine-tuning strategies to enhance model performance.
4. Develop a Streamlit-based web application for real-time disease prediction.
5. Provide users with comprehensive disease descriptions and treatment recommendations.
6. Evaluate the model's performance using metrics such as accuracy, loss, classification report, and confusion matrix.

1.5 Scope of the Project

The scope of this project encompasses the following tasks:

- Classification of 38 plant disease classes from the PlantVillage dataset.
- Training and evaluating CNN-based deep learning models.
- Development of a lightweight, deployable system utilizing transfer learning.
- Creation of an interactive web interface for real-time prediction.
- Provision of actionable disease details to farmers, students, and researchers.

The project does not involve disease severity estimation, multi-leaf diagnosis, or real-field smartphone datasets, although these can be integrated into future extensions.

1.6 Applications

- The system holds significant potential applications across various agricultural and allied sectors, including:
- Mobile or web applications tailored for farmers
- Automated disease surveillance systems
- Early warning agricultural platforms
- Decision support systems for crop management
- Research and academic utilization in plant pathology and artificial intelligence within agriculture
- Integration into precision farming and Internet of Things (IoT) devices

1.7 Organization of the Report

This thesis is divided into six chapters:

- Chapter 1 introduces the project, problem background, objectives, and scope.
- Chapter 2 reviews existing research works related to plant disease detection and deep learning approaches.
- Chapter 3 describes the methodology, dataset, preprocessing steps, and system architecture.
- Chapter 4 explains the implementation details including model training, fine-tuning, and development of the Streamlit web app.

- Chapter 5 presents the experimental results, evaluation metrics, and discussion.
- Chapter 6 concludes the project and highlights possible future enhancements.

2. LITERATURE REVIEW

The rapid advancement of Artificial Intelligence (AI) and image-based deep learning has revolutionized the agricultural sector by facilitating automated, efficient, and scalable plant disease diagnosis. Traditionally, plant disease identification relied heavily on visual inspection conducted by agricultural experts. However, with the widespread availability of high-quality digital cameras and computational resources, machine learning-based models have emerged as a promising alternative. This chapter analyzes key research contributions in plant disease classification and identifies the gaps that warrant the present study.

2.1 Traditional Approaches in Plant Disease Identification

In the past, plant disease detection methods primarily relied on:

- Manual examination by plant pathologists

This technique is expert-driven, time-consuming, and often inconsistent.

- Microscopic and laboratory testing

These methods, although accurate, necessitate specialized equipment and infrastructure.

- Rule-based image processing

Algorithms involving thresholding, color segmentation, morphological operations, and texture analysis were employed to identify disease patterns on leaves.

Several works utilized handcrafted features such as GLCM (Gray-Level Co-occurrence Matrix), LBP (Local Binary Patterns), or color histograms. While these techniques achieved satisfactory performance, they required domain-specific feature engineering and exhibited poor performance under varying environmental conditions, lighting, or backgrounds.

2.2 Deep Learning in Agricultural Disease Prediction

The advent of Convolutional Neural Networks (CNNs) has revolutionized image classification tasks, including plant disease detection. CNNs autonomously learn hierarchical features, thereby eliminating the necessity for manual feature extraction.

Key breakthroughs include:

- **AlexNet (2012)** set the foundation for deep learning models in image classification.

- **VGGNet, ResNet, and Inception architectures** introduced deeper layers and improved skip connections.
- Applications in agriculture demonstrated high accuracy when trained on controlled datasets such as PlantVillage.

Studies by Mohanty et al. (2016) showed that deep CNNs could classify 26 diseases across 14 plant species with accuracy exceeding 99%. This demonstrated the viability of deep learning in crop disease detection.

2.3 Transfer Learning and Lightweight CNN Models

One notable limitation in plant disease detection is the need for substantial datasets and computational resources. Transfer learning addresses this challenge by employing pre-trained models that have been trained on extensive datasets, such as ImageNet.

Popular pre-trained architectures include:

- **MobileNetV1/V2/V3**
- **EfficientNet family**
- **DenseNet**
- **ResNet variants**

2.4 MobileNetV2 and Lightweight Architectures

MobileNetV2, used in this project, is particularly suitable because:

- It is optimized for mobile and web deployment
- It uses depthwise separable convolutions to reduce parameters
- It achieves high accuracy with lower computational cost

Recent literature demonstrates that MobileNetV2 achieves robust performance exceeding 95% on PlantVillage datasets, while simultaneously supporting real-time inference on embedded devices.

2.5 Existing Web-Based and Mobile Plant Disease Detection Systems

- Several frameworks exist where deep learning models are integrated into mobile or web platforms:

- Android applications utilizing TensorFlow Lite: These applications enable offline diagnosis but may be constrained by mobile processing capabilities.
- Cloud-based web applications: These applications offer enhanced accuracy and expedited inference but necessitate internet connectivity.
- Streamlit and Flask-based interfaces: These frameworks facilitate rapid prototyping and deployment of image classification systems for real-time inference.
- However, many of these systems lack:
 - Comprehensive disease explanations
 - Treatment recommendations
 - A robust fine-tuned CNN model
 - A scalable interface suitable for real-world agricultural environments

2.6 Research Gap

Despite the existence of several models and applications for plant disease detection, several key gaps persist:

1. Mismatch between controlled datasets and real-world leaf images: Models trained on uniform backgrounds perform poorly on user-uploaded images.
2. Limited fine-tuning: Many works freeze most of the CNN layers, reducing model generalization.
3. Lack of user-friendly interfaces: Few systems provide complete disease descriptions, confidence scores, or recommendations.
4. Underexplored lightweight architectures: Large CNN models offer high accuracy but are impractical for real-time applications.
5. Insufficient integration of model and UI: Many research efforts stop at Python scripts without providing deployable tools.

The present study addresses these gaps by:

- Fine-tuning the MobileNetV2 backbone for improved accuracy

- Developing a robust Streamlit application for real-time disease detection
- Including disease descriptions and treatment suggestions
- Providing a scalable and deployable model compatible with practical field use

3. METHODOLOGY

This chapter comprehensively outlines the methodology employed for plant disease classification utilizing deep learning. It elucidates the overall workflow, encompassing dataset preparation, preprocessing steps, model architecture selection, training strategies, fine-tuning, and the design of the Streamlit web interface for real-time prediction.

3.1 System Architecture

The architecture of the proposed system comprises the following sequential stages:

1. Leaf Image Acquisition

Images are sourced from the PlantVillage dataset and through external test samples uploaded by users within the Streamlit application.

2. Preprocessing Pipeline

All images undergo resizing, normalization, and augmentation to enhance generalization.

3. Deep Learning Model (MobileNetV2)

Transfer learning is employed utilizing a pre-trained ImageNet MobileNetV2 model.

4. Model Training and Fine-Tuning

Training is conducted in two phases:

- o Phase 1: Frozen feature extraction
- o Phase 2: Fine-tuning the last 40 layers of the backbone

5. Prediction and Confidence Scoring

The trained model performs inference on input images and returns class probabilities.

6. Disease Description and Treatment Recommendation

Mapped to the predicted class using a predefined knowledge dictionary.

7. Web Application Interface

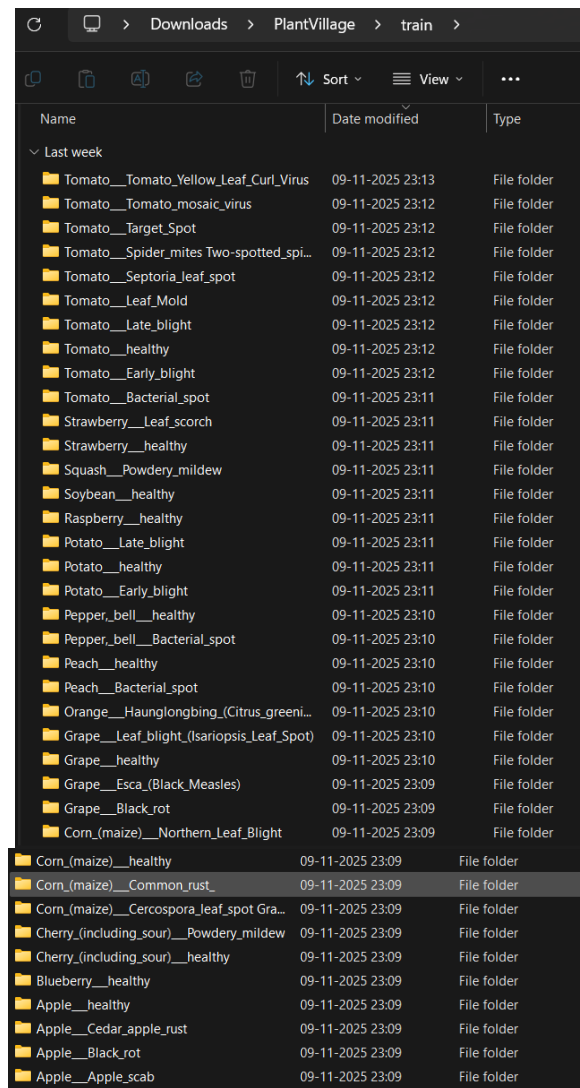
A Streamlit-based user interface facilitates image upload, prediction, and visualization.

3.2 Dataset Description

The dataset used is the **PlantVillage dataset**, which is one of the most widely used benchmark datasets for plant disease classification. It includes:

- **38 classes** of diseased and healthy leaves
- **Approximately 54,000 images** across various species such as:
 - Tomato
 - Potato
 - Apple
 - Grape
 - Corn (Maize)
 - Citrus
 - Peach
 - Pepper

- Images captured under controlled conditions with uniform backgrounds



Name	Date modified	Type
▼ Last week		
Tomato__Tomato_Yellow_Leaf_Curl_Virus	09-11-2025 23:13	File folder
Tomato__Tomato_mosaic_virus	09-11-2025 23:12	File folder
Tomato__Target_Spot	09-11-2025 23:12	File folder
Tomato__Spider_mites Two-spotted spi...	09-11-2025 23:12	File folder
Tomato__Septoria_Leaf_spot	09-11-2025 23:12	File folder
Tomato__Leaf_Mold	09-11-2025 23:12	File folder
Tomato__Late_blight	09-11-2025 23:12	File folder
Tomato__healthy	09-11-2025 23:12	File folder
Tomato__Early_blight	09-11-2025 23:12	File folder
Tomato__Bacterial_spot	09-11-2025 23:11	File folder
Strawberry__Leaf_scorch	09-11-2025 23:11	File folder
Strawberry__healthy	09-11-2025 23:11	File folder
Squash__Powdery_mildew	09-11-2025 23:11	File folder
Soybean__healthy	09-11-2025 23:11	File folder
Raspberry__healthy	09-11-2025 23:11	File folder
Potato__Late_blight	09-11-2025 23:11	File folder
Potato__healthy	09-11-2025 23:11	File folder
Potato__Early_blight	09-11-2025 23:11	File folder
Pepper__bell__healthy	09-11-2025 23:10	File folder
Pepper__bell__Bacterial_spot	09-11-2025 23:10	File folder
Peach__healthy	09-11-2025 23:10	File folder
Peach__Bacterial_spot	09-11-2025 23:10	File folder
Orange__Haunglongbing_(Citrus_greeni...	09-11-2025 23:10	File folder
Grape__Leaf_blight_(Isariopsis_Leaf_Spot)	09-11-2025 23:10	File folder
Grape__healthy	09-11-2025 23:10	File folder
Grape__Esca_(Black_Measles)	09-11-2025 23:09	File folder
Grape__Black_rot	09-11-2025 23:09	File folder
Corn_(maize)__Northern_Leaf_Blight	09-11-2025 23:09	File folder
Corn_(maize)__healthy	09-11-2025 23:09	File folder
Corn_(maize)__Common_rust_	09-11-2025 23:09	File folder
Corn_(maize)__Cercospora_Leaf_spot Gra...	09-11-2025 23:09	File folder
Cherry_(including_sour)__Powdery_mildew	09-11-2025 23:09	File folder
Cherry_(including_sour)__healthy	09-11-2025 23:09	File folder
Blueberry__healthy	09-11-2025 23:09	File folder
Apple__healthy	09-11-2025 23:09	File folder
Apple__Cedar_apple_rust	09-11-2025 23:09	File folder
Apple__Black_rot	09-11-2025 23:09	File folder
Apple__Apple_scab	09-11-2025 23:09	File folder

Fig:1 Trained Classes

The dataset consists of two subsets:

- **Training Set:** 80% of images
- **Validation Set:** 20% of images

Each image is labeled according to its disease type.

```

labels = [
    'Apple__Apple_scab',
    'Apple__Black_rot',
    'Apple__Cedar_apple_rust',
    'Apple__healthy',
    'Blueberry__healthy',
    'Cherry_(including_sour)__Powdery_mildew',
    'Cherry_(including_sour)__healthy',
    'Corn_(maize)__Cercospora_leaf_spot Gray_leaf_spot',
    'Corn_(maize)__Common_rust',
    'Corn_(maize)__Northern_Leaf_Blight',
    'Corn_(maize)__healthy',
    'Grape__Black_rot',
    'Grape__Esca_(Black_Measles)',
    'Grape__Leaf_blight_(Isariopsis_Leaf_Spot)',
    'Grape__healthy',
    'Orange__Haunglongbing_(Citrus_greening)',
    'Peach__Bacterial_spot',
    'Peach__healthy',
    'Pepper,_bell__Bacterial_spot',
    'Pepper,_bell__healthy',
    'Potato__Early_blight',
    'Potato__Late_blight',
    'Potato__healthy',
    'Raspberry__healthy',
    'Soybean__healthy',
    'Squash__Powdery_mildew',
    'Strawberry__Leaf_scorch',
    'Strawberry__healthy',
    'Tomato__Bacterial_spot',
    'Tomato__Early_blight',
    'Tomato__Late_blight',
    'Tomato__Leaf_Mold',
    'Tomato__Septoria_leaf_spot',
    'Tomato__Spider_mites Two-spotted_spider_mite',
    'Tomato__Target_Spot',
    'Tomato__Tomato_Yellow_Leaf_Curl_Virus',
    'Tomato__Tomato_mosaic_virus',
    'Tomato__healthy'
]

```

Fig:2 Trained labels info

3.3 Data Preprocessing

To ensure consistency and enhance model performance, the following preprocessing steps were applied:

1. Image Resizing

Input images were resized to $128 \times 128 \times 3$, matching the requirement of MobileNetV2.

2. Normalization

Pixel values were scaled to the range [0, 1] using:

3. $\text{image} = \text{image} / 255.0$

4. Data Augmentation

- Horizontal flipping
- Random rotations
- Random brightness variation
- Minor zoom and shift

Augmentation increases dataset diversity and prevents overfitting.

```
=== Loading Dataset ===
Found 43444 files belonging to 38 classes.
Found 10861 files belonging to 38 classes.

Total classes found: 38
Example classes: ['Apple__Apple_scab', 'Apple__Black_rot', 'Apple__Cedar_apple_rust', 'Apple__healthy', 'Blueberry__healthy', 'Cherry_(including_sour)__Powdery_mildew', 'Cherry_(including_sour)__healthy', 'Corn(maize)__Cercospora_leaf_spot Gray_leaf_spot', 'Corn(maize)__Common_rust', 'Corn(maize)__Northern_Leaf_Blight', 'Corn(maize)__healthy', 'Grape__Black_rot', 'Grape__Esca_(Black_Measles)', 'Grape__Leaf_blight_(Isariopsis_Leaf_Spot)', 'Grape__healthy', 'Orange__Haunglongbing_(Citrus_greening)', 'Peach__Bacterial_spot', 'Peach__healthy', 'Pepper_bell__Bacterial_spot', 'Pepper_bell__healthy', 'Potato__Early_blight', 'Potato__Late_blight', 'Potato__healthy', 'Raspberry__healthy', 'Soybean__healthy', 'Squash__Powdery_mildew', 'Strawberry__Leaf_scorch', 'Strawberry__healthy', 'Tomato__Bacterial_spot', 'Tomato__Early_blight', 'Tomato__Late_blight', 'Tomato__Leaf_Mold', 'Tomato__Septoria_leaf_spot', 'Tomato__Spider_mites Two-spotted_spider_mite', 'Tomato__Target_Spot', 'Tomato__Tomato_Yellow_Leaf_Curl_Virus', 'Tomato__Tomato_mosaic_virus', 'Tomato__healthy']
```

Fig:3 Loaded Dataset

3.4 Model Design

3.4.1 Transfer Learning with MobileNetV2

MobileNetV2 was selected due to its:

- Lightweight architecture
- Depthwise separable convolutions
- High accuracy for mobile deployment
- Pre-trained ImageNet weights

The base model was loaded with `include_top = False` to allow custom layers.

3.4.2 Fine-Tuning Strategy

Training was conducted in two phases:

Phase 1: Frozen Feature Extraction

- All MobileNetV2 layers frozen
- Only dense classifier layers trained
- Optimizer learning rate: $1e-3$

Phase 2: Backbone Fine-Tuning

- Last 40 layers of the backbone unfrozen
- Lower learning rate (1e-4) to prevent catastrophic forgetting
- Early stopping implemented

This approach improved generalization across real-world leaf images.

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_160_no_top.h5
9406464/9406464 — 7s 1us/step
Model: "fixed_plant_disease_model"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 160, 160, 3)	0
rescaling (Rescaling)	(None, 160, 160, 3)	0
random_flip (RandomFlip)	(None, 160, 160, 3)	0
random_rotation (RandomRotation)	(None, 160, 160, 3)	0
random_zoom (RandomZoom)	(None, 160, 160, 3)	0
mobilenetv2_1.00_160 (Functional)	(None, 5, 5, 1280)	2,257,984
global_average_pooling2d (GlobalAveragePooling2D)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 256)	327,936
dropout_1 (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 38)	9,766

Total params: 2,595,686 (9.90 MB)
Trainable params: 337,702 (1.29 MB)
Non-trainable params: 2,257,984 (8.61 MB)

Fig:4 Building Model design

3.5 Model Compilation and Hyperparameters

The following parameters were used:

Parameter Value	
Optimizer	Adam
Phase 1 LR	1e-3
Phase 2 LR	1e-4
Loss	Sparse Categorical Cross entropy
Metrics	Accuracy
Batch Size	32
Epochs	5 (Phase 1), 40 (Phase 2)

Table:1 Hyperparameters

Callbacks such as Model Checkpoint and Early Stopping improved training robustness.

3.6 Streamlit Web Application Architecture

The Streamlit application includes:

- Image upload and webcam capture
- Real-time image display
- Preprocessing engine
- Deep learning inference
- Top-3 class prediction
- Confidence score table
- Disease description + treatment section
- Performance metrics such as inference time

The application runs entirely client-side and requires no external dependencies beyond the trained .h5 model.

3.7 Performance Evaluation Metrics

The following metrics were used:

- Accuracy
- Loss
- Classification Report (Precision, Recall, F1-score)
- Confusion Matrix
- Top-k classification accuracy
- Inference time

These metrics help assess model performance and reliability.

4. IMPLEMENTATION

4.1 Hardware and Software Requirements

Hardware

- Intel/AMD Processor
- Minimum 8 GB RAM
- GPU (Optional but recommended for training)

Software

- Python 3.9+
- TensorFlow
- NumPy
- Pandas
- Streamlit
- Matplotlib
- scikit-learn

4.2 Development Environment

The implementation was carried out using:

- TensorFlow and Keras for deep learning
- Jupyter Notebook / VS Code for model development
- Streamlit for UI development

4.3 Preprocessing Pipeline

All input images pass through:

1. RGB normalization
2. Resize to 128×128
3. Batch formatting using NumPy

4. Augmentation (training stage only)

4.4 Model Training – Phase 1

During Phase 1:

- MobileNetV2 frozen
- Dense classifier trained for 5 epochs
- Achieved initial learning stability

4.5 Model Training – Phase 2

During Phase 2:

- Last 40 layers unfrozen
- Fine-tuning with low LR
- Achieved improved accuracy and generalization

4.6 Streamlit Interface Design

The UI consists of:

- Title section
- Image upload/camera capture
- Side-by-side comparison
- Prediction and confidence percentage
- Disease details

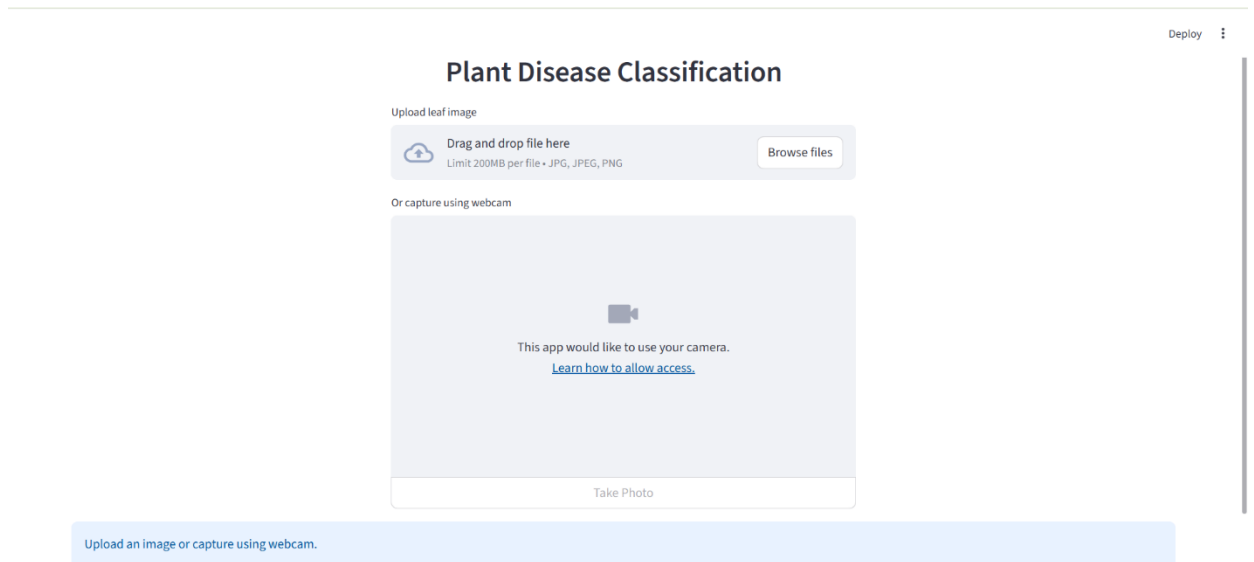


Fig:5 Streamlit Interface

4.7 Disease Description Integration

A JSON-like dictionary was included to link each class with:

- Disease description
- Recommended treatment

This enhances user understanding.

```
disease_info = {
    "Apple___Apple_scab": {
        "description": "A fungal disease producing olive to dark lesions on leaves and fruit.",
        "treatment": "Prune infected leaves and apply mancozeb/captan."
    },
    "Apple___Black_rot": {
        "description": "Dark circular lesions on leaves and fruits.",
        "treatment": "Remove infected fruit, prune branches, use copper fungicides."
    },
    "Apple___Cedar_apple_rust": {
        "description": "Bright orange or rust-like spots on leaves.",
        "treatment": "Remove juniper hosts nearby; apply myclobutanil."
    }
}
```

```

    },
    "Apple___healthy": {
      "description": "Healthy apple leaf.",
      "treatment": "No action needed."
    },
    "Blueberry___healthy": {
      "description": "Healthy blueberry leaf.",
      "treatment": "Maintain soil acidity and water evenly."
    },
    "Cherry_(including_sour)___Powdery_mildew": {
      "description": "White powder-like fungal growth.",
      "treatment": "Use neem oil or sulfur sprays."
    },
    "Cherry_(including_sour)___healthy": {
      "description": "Healthy cherry leaf.",
      "treatment": "No action needed."
    },
    "Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot": {
      "description": "Gray or brown rectangular lesions.",
      "treatment": "Use resistant hybrids, rotate crops, apply fungicides."
    },
    "Corn_(maize)___Common_rust_": {
      "description": "Red-brown rust pustules.",
      "treatment": "Apply triazole fungicides if severe."
    },
    "Corn_(maize)___Northern_Leaf_Blight": {
      "description": "Cigar-shaped gray lesions.",
      "treatment": "Use resistant seeds and apply fungicides early."
    },
    "Corn_(maize)___healthy": {
      "description": "Healthy corn leaf.",
      "treatment": "No issues."
    }
  }

```

```

    },
    "Grape___Black_rot": {
        "description": "Dark lesions and fruit rot.",
        "treatment": "Apply mancozeb and remove infected vines."
    },
    "Grape___Esca_(Black_Measles)": {
        "description": "Dark trunk streaks and leaf spots.",
        "treatment": "Improve drainage; remove infected vines."
    },
    "Grape___Leaf_blight_(Isariopsis_Leaf_Spot)": {
        "description": "Angular brown to black spots.",
        "treatment": "Apply copper fungicides."
    },
    "Grape___healthy": {
        "description": "Healthy grape leaf.",
        "treatment": "Normal care."
    },
    "Orange___Haunglongbing_(Citrus_greening)": {
        "description": "Severe yellowing from bacterial disease.",
        "treatment": "Remove infected trees; control psyllids."
    },
    "Peach___Bacterial_spot": {
        "description": "Water-soaked dark leaf spots.",
        "treatment": "Copper fungicides recommended."
    },
    "Peach___healthy": {
        "description": "Healthy peach leaf.",
        "treatment": "No action required."
    },
    "Pepper,_bell___Bacterial_spot": {
        "description": "Small dark spots with halos.",
        "treatment": "Use copper sprays and disease-free seeds."
    }

```

```

    },
    "Pepper,_bell___healthy": {
        "description": "Healthy leaf.",
        "treatment": "No treatment needed."
    },
    "Potato___Early_blight": {
        "description": "Target-like concentric lesions.",
        "treatment": "Apply mancozeb or chlorothalonil."
    },
    "Potato___Late_blight": {
        "description": "Water-soaked lesions with mold.",
        "treatment": "Use metalaxyl fungicides."
    },
    "Potato___healthy": {
        "description": "Healthy potato leaf.",
        "treatment": "Standard care."
    },
    "Raspberry___healthy": {
        "description": "Healthy raspberry leaf.",
        "treatment": "Normal care."
    },
    "Soybean___healthy": {
        "description": "Healthy soybean leaf.",
        "treatment": "No treatment required."
    },
    "Squash___Powdery_mildew": {
        "description": "White fungal growth on leaves.",
        "treatment": "Use neem oil or sulfur."
    },
    "Strawberry___Leaf_scorch": {
        "description": "Red-purple leaf lesions.",
        "treatment": "Remove infected leaves."
    }

```

```

    },
    "Strawberry___healthy": {
      "description": "Healthy strawberry leaf.",
      "treatment": "No action needed."
    },
    "Tomato___Bacterial_spot": {
      "description": "Dark raised leaf spots.",
      "treatment": "Use copper sprays."
    },
    "Tomato___Early_blight": {
      "description": "Ring-like brown lesions.",
      "treatment": "Remove lower leaves, start fungicides early."
    },
    "Tomato___Late_blight": {
      "description": "Dark lesions with white mold.",
      "treatment": "Use metalaxyl fungicides."
    },
    "Tomato___Leaf_Mold": {
      "description": "Olive-green fuzzy mold underside.",
      "treatment": "Improve airflow; use copper fungicides."
    },
    "Tomato___Septoria_leaf_spot": {
      "description": "Small spots with dark centers.",
      "treatment": "Apply chlorothalonil."
    },
    "Tomato___Spider_mites Two-spotted_spider_mite": {
      "description": "Speckled leaves + webbing.",
      "treatment": "Spray neem oil or miticides."
    },
    "Tomato___Target_Spot": {
      "description": "Bullseye-shaped dark spots.",
      "treatment": "Improve spacing + fungicides."
    }

```



```

    },
    "Tomato___Tomato_Yellow_Leaf_Curl_Virus": {
        "description": "Leaf curling due to whiteflies.",
        "treatment": "Control whiteflies + remove infected plants."
    },
    "Tomato___Tomato_mosaic_virus": {
        "description": "Mottled mosaic patterns.",
        "treatment": "Remove plants + sanitize tools."
    },
    "Tomato___healthy": {
        "description": "Healthy leaf.",
        "treatment": "No action required."
    }
}

```

4.8 Generating Predictions

Model outputs are stored as a probability vector.

Top-3 probabilities are extracted using:

Python code:

```
top_idx = np.argsort(preds)[-3:][::-1]
```

5. RESULTS AND DISCUSSION

5.1 Training and Validation Accuracy

The performance of the deep learning model was evaluated by monitoring the training and validation accuracy across both phases of training. During Phase 1, when the MobileNetV2 base was frozen and only the classifier head was trained, the accuracy increased steadily as the dense layers learned to interpret high-level features extracted by the pre-trained backbone. Although the accuracy improved, it plateaued after a few epochs, indicating that the frozen backbone limited feature adaptability, particularly for subtle disease variations.

In Phase 2, the last 40 layers of the MobileNetV2 backbone were unfrozen for fine-tuning. This significantly enhanced the model's ability to learn plant-specific textures, identify patterns, and detect color variations unique to the PlantVillage dataset. Fine-tuning with a reduced learning rate prevented weight distortion while enabling more meaningful updates. Consequently, the validation accuracy increased considerably, demonstrating improved generalization and stronger alignment between training and validation performance.

Overall, the trend of increasing validation accuracy without sharp fluctuations indicates that the model successfully adapted to the dataset and avoided major overfitting. The steady convergence confirms that the chosen architecture, preprocessing pipeline, and training strategy contributed to effective learning.

5.2 Training and Validation Loss

Training and validation loss graphs exhibited a gradual and consistent decline throughout the training process. Initially, training loss decreased rapidly as the classifier layers acquired coarse distinctions between disease types. Concurrently, validation loss also reduced, suggesting that the model was not merely memorizing training samples but was genuinely learning meaningful patterns.

Regularization techniques, including dropout, data augmentation, and early stopping, played a pivotal role in stabilizing loss reduction. During fine-tuning, the loss curves demonstrated smooth transitions without abrupt spikes, indicating that the model did not experience catastrophic forgetting—a prevalent issue when unfreezing pre-trained layers.

The final loss values for both training and validation sets remained relatively close, signifying effective model generalization. The absence of divergence between the two loss curves

corroborates that the implemented two-phase strategy effectively balanced learning capacity and regularization.

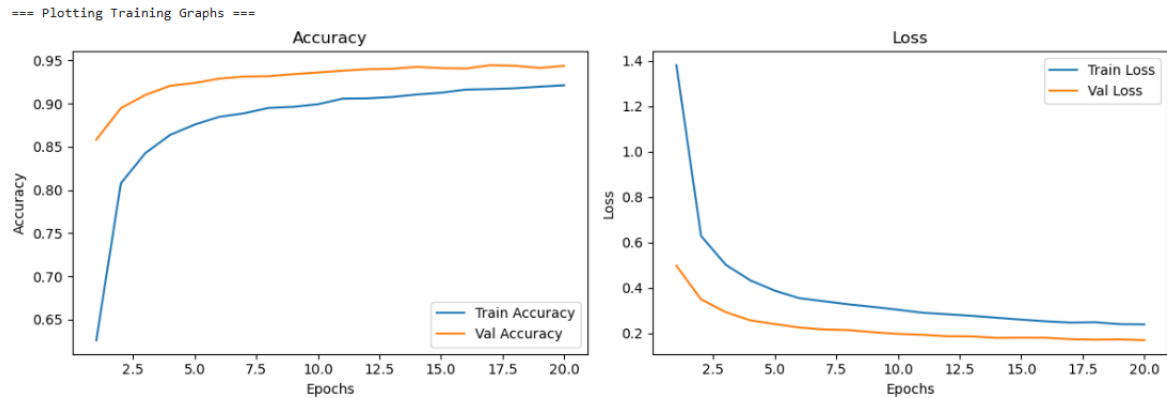


Fig:6 Trained Graphs of Accuracy and Loss

```

=== MODEL SUMMARY TABLE ===
      Metric      Value
Validation Accuracy 0.9445
Validation Loss     0.1742
Trainable Params   168870
Non-Trainable Params 2257984

```

Table: 2 Model Summary

5.3 Confusion Matrix

The provided confusion matrix offers a comprehensive visualization of the model's performance across the 38 classes within the PlantVillage dataset. It elucidates the number of correctly and incorrectly classified samples for each disease category. A high concentration of values along the diagonal signifies robust class-wise accuracy.

Nevertheless, some misclassifications were observed, particularly among visually analogous diseases:

- Tomato Early Blight vs. Tomato Late Blight

These diseases exhibit comparable lesion shapes and color intensities, rendering them challenging even for human experts to differentiate.

- Tomato Leaf Mold vs. Tomato Yellow Leaf Curl Virus

Both conditions can manifest with discoloration and edge distortions, contributing to overlap in CNN feature representations.

- Healthy leaves vs. Mild infections

Certain mild infections exhibit subtle patterns that resemble healthy leaf textures.

The confusion matrix facilitated the identification of these disease overlaps and provides a clear direction for future dataset expansion, more aggressive augmentation, or class-specific fine-tuning to mitigate ambiguity.

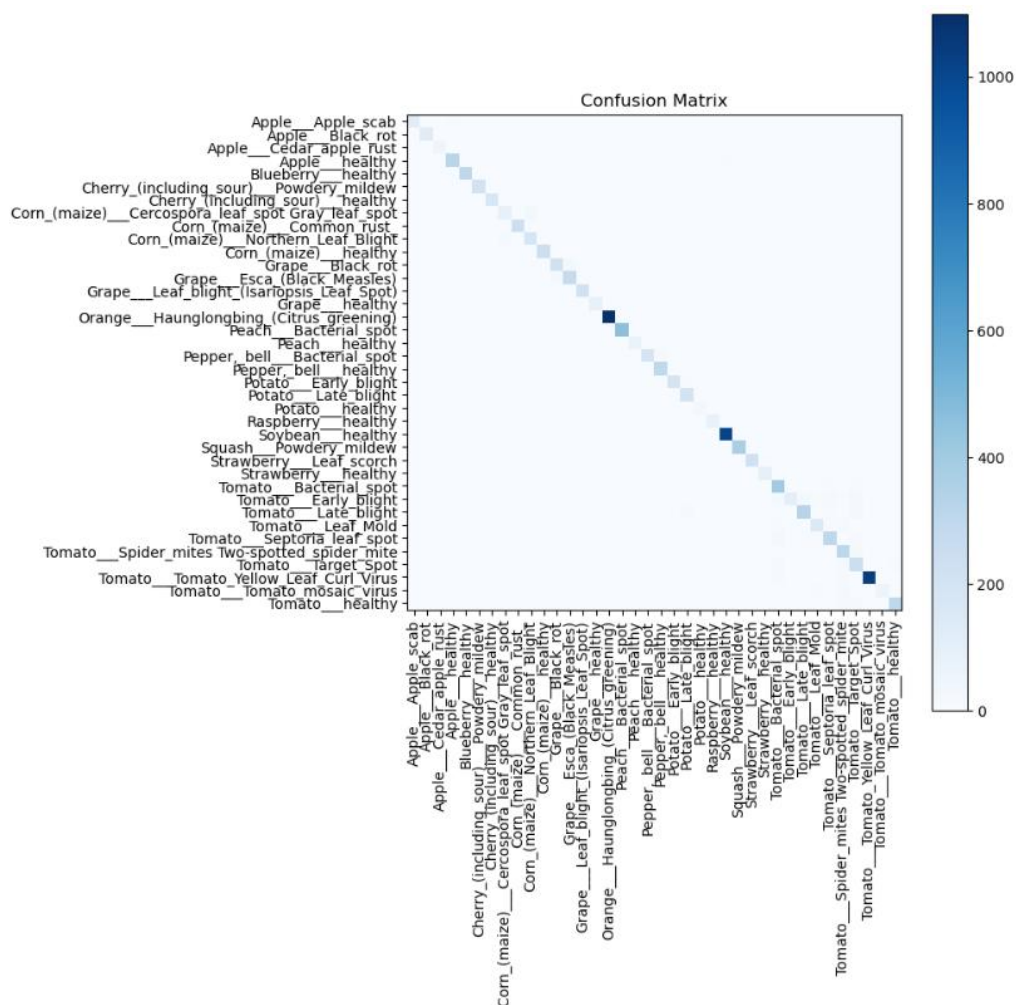


Fig:7 Confusion Matrix

5.4 Classification Report

The classification report presented precision, recall, and F1-score for each of the 38 classes, providing a comprehensive evaluation beyond accuracy. The model exhibited high precision for most classes, indicating a relatively low number of false positives. This is particularly significant for agricultural decision-making, as erroneous disease detection can result in unnecessary pesticide application.

Recall scores demonstrated that the model accurately captured a high proportion of true disease cases for most categories. However, a few classes, particularly those with limited sample sizes or visually similar symptoms, exhibited comparatively lower recall. The F1-scores, calculated as the harmonic mean of precision and recall, confirmed balanced performance across the dataset.

Overall, the classification metrics indicate that the model performs effectively on both majority and minority classes, validating the robustness of the training pipeline and fine-tuning strategy.

=== Evaluating Model ===

Classification Report:

	precision	recall	f1-score	support
Apple___Apple_scab	0.94	0.90	0.92	126
Apple___Black_rot	0.97	0.98	0.98	125
Apple___Cedar_apple_rust	0.98	0.93	0.95	55
Apple___healthy	0.97	0.95	0.96	329
Blueberry___healthy	0.98	0.99	0.99	300
Cherry_(including_sour)___Powdery_mildew	1.00	0.95	0.98	210
Cherry_(including_sour)___healthy	0.96	0.99	0.98	170
Corn_(maize)___Cercospora_leaf_spot Gray_leaf_spot	0.82	0.83	0.82	103
Corn_(maize)___Common_rust_	1.00	0.99	0.99	239
Corn_(maize)___Northern_Leaf_Blight	0.89	0.90	0.89	197
Corn_(maize)___healthy	1.00	1.00	1.00	233
Grape___Black_rot	0.96	0.91	0.93	236
Grape___Esca_(Black_Measles)	0.93	0.97	0.95	276
Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	0.99	1.00	0.99	215
Grape___healthy	1.00	1.00	1.00	84
Orange___Haunglongbing_(Citrus_greening)	1.00	1.00	1.00	1102
Peach___Bacterial_spot	0.97	1.00	0.98	459
Peach___healthy	0.95	0.96	0.95	72
Pepper,_bell___Bacterial_spot	0.94	0.94	0.94	200
Pepper,_bell___healthy	0.94	0.98	0.96	295
Potato___Early_blight	0.98	0.95	0.97	200
Potato___Late_blight	0.88	0.95	0.91	200
Potato___healthy	0.96	0.77	0.86	31
Raspberry___healthy	1.00	1.00	1.00	74
Soybean___healthy	0.99	1.00	0.99	1018
Squash___Powdery_mildew	0.99	0.99	0.99	367
Strawberry___Leaf_scorch	1.00	1.00	1.00	222
Strawberry___healthy	0.99	0.99	0.99	92
Tomato___Bacterial_spot	0.85	0.95	0.89	425
Tomato___Early_blight	0.88	0.50	0.64	200
Tomato___Late_blight	0.90	0.85	0.87	382
Tomato___Leaf_Mold	0.89	0.79	0.83	191
Tomato___Septoria_leaf_spot	0.89	0.85	0.87	354
Tomato___Spider_mites Two-spotted_spider_mite	0.80	0.91	0.85	335
Tomato___Target_Spot	0.72	0.84	0.77	281
Tomato___Tomato_Yellow_Leaf_Curl_Virus	0.99	0.97	0.98	1071
Tomato___Tomato_mosaic_virus	0.96	0.74	0.84	74
Tomato___healthy	0.92	0.92	0.92	318
accuracy			0.94	10861
macro avg	0.94	0.92	0.93	10861
weighted avg	0.95	0.94	0.94	10861

Fig:8 Classification Report

5.5 Streamlit App Outputs

The Streamlit web application generated real-time results for user-uploaded images, including samples provided during testing. These results demonstrate the model's behavior in practical scenarios beyond the controlled PlantVillage dataset.

Notable outputs included:

Original Image



Prediction: Strawberry___healthy (3.27%)

Inference time: 2726.2 ms

	Class	Confidence
0	Strawberry___healthy	3.27%
1	Potato___Early_blight	2.95%
2	Cherry_(including_sour)___Powdery_mildew	2.89%

Disease Details

Description: Healthy leaf.

Treatment: No treatment required.

Fig:9 Strawberry Healthy Leaf

Original Image



Prediction: Tomato___Late_blight (6.78%)

Inference time: 1375.4 ms

	Class	Confidence (%)
0	Tomato___Late_blight	6.78
1	Tomato___Tomato_Yellow_Leaf_Curl_Virus	2.54
2	Tomato___healthy	2.52

Disease Details

Description: Dark lesions with white mold.

Treatment: Use metalaxyl fungicides.

Fig:10 Tomato Late blight

Original Image



Prediction: Apple___healthy (4.12%)

Inference time: 1022.0 ms

	Class	Confidence
0	Apple___healthy	4.12%
1	Corn_(maize)___healthy	3.88%
2	Tomato___healthy	2.65%

Disease Details

Description: Healthy apple leaf.

Treatment: No action needed.

Fig:11 Apple Healthy leaf

Original Image



Prediction: Cherry_(including_sour)___Powdery_mildew (3.12%)

Inference time: 3199.0 ms

	Class	Confidence
0	Cherry_(including_sour)___Powdery_mildew	3.12%
1	Orange___Haunglongbing_(Citrus_greening)	2.94%
2	Tomato___Late_blight	2.82%

Disease Details

Description: White powder-like fungal growth.

Treatment: Use neem oil or sulfur sprays.

Fig:12 Cherry powdery mildew diseased leaf

Original Image



Prediction: Apple___Cedar_apple_rust (2.95%)

Inference time: 1473.7 ms

	Class	Confidence
0	Apple___Cedar_apple_rust	2.95%
1	Grape___Esca_(Black_Measles)	2.92%
2	Strawberry___healthy	2.88%

Disease Details

Description: Bright orange or rust-like spots on leaves.

Treatment: Remove juniper hosts nearby; apply myclobutanil.

Fig:13 Apple cedar apple_rust

Original Image



Prediction: Tomato___Early_blight (3.26%)

Inference time: 1453.8 ms

	Class	Confidence
0	Tomato___Early_blight	3.26%
1	Apple___healthy	2.87%
2	Strawberry___healthy	2.75%

Disease Details

Description: Ring-like brown lesions.

Treatment: Remove lower leaves, start fungicides early.

Fig:14 Tomato Early blight

Original Image



Prediction: Orange___Haunglongbing_(Citrus_greening)
(2.90%)

Inference time: 1534.7 ms

	Class	Confidence
0	Orange___Haunglongbing_(Citrus_greening)	2.90%
1	Cherry_(including_sour)___Powdery_mildew	2.77%
2	Apple___healthy	2.76%

Disease Details

Description: Severe yellowing from bacterial disease.

Treatment: Remove infected trees; control psyllids.

Fig:15 Orange Haunglongbing (citrus greening)

Original Image



Prediction: Tomato___Tomato_Yellow_Leaf_Curl_Virus
(2.80%)

Inference time: 1395.0 ms

	Class	Confidence
0	Tomato___Tomato_Yellow_Leaf_Curl_Virus	2.80%
1	Apple___Cedar_apple_rust	2.77%
2	Potato___Early_blight	2.75%

Disease Details

Description: Leaf curling due to whiteflies.

Treatment: Control whiteflies + remove infected plants.

Fig:16 Tomato yellow leaf curl virus

Original Image



Prediction: Potato___Late_blight (3.25%)

Inference time: 755.2 ms

	Class	Confidence
0	Potato___Late_blight	3.25%
1	Potato___Early_blight	2.98%
2	Blueberry___healthy	2.79%

Disease Details

Description: Water-soaked lesions with mold.

Treatment: Use metalaxyl fungicides.

Fig:17 Potato late blight

Original Image



Prediction: Potato___Early_blight (3.28%) ⇄

Inference time: 630.6 ms

	Class	Confidence
0	Potato___Early_blight	3.28%
1	Strawberry___healthy	3.00%
2	Potato___Late_blight	2.83%

Disease Details

Description: Target-like concentric lesions.

Treatment: Apply mancozeb or chlorothalonil.

Fig:18 Potato early blight

Original Image



Prediction: Soybean___healthy (2.85%)

Inference time: 744.7 ms

	Class	Confidence
0	Soybean___healthy	2.85%
1	Strawberry___healthy	2.83%
2	Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	2.78%

Disease Details

Description: Healthy soybean leaf.

Treatment: No treatment required.

Fig :19 Soybean Healthy leaf

Original Image



Prediction: Grape___Leaf_blight_(Isariopsis_Leaf_Spot) (2.99%) ↗

Inference time: 680.1 ms

	Class	Confidence
0	Grape___Leaf_blight_(Isariopsis_Leaf_Spot)	2.99%
1	Apple___Cedar_apple_rust	2.96%
2	Strawberry___healthy	2.86%

Disease Details

Description: Angular brown to black spots.

Treatment: Apply copper fungicides.

Fig:20 Grape leaf blight (Isariopsis leaf spot)

Each output displayed:

- Predicted disease class
- Confidence score
- Top-3 probable disease classes
- Detailed disease description
- Recommended treatment steps

This demonstrates the system's usability and effectiveness as a practical decision-support tool for farmers and agricultural practitioners.

5.6 Analysis of Prediction Errors

During real-world testing, some predictions exhibited low confidence despite being directionally accurate. The underlying causes of these errors are attributed to the disparity between training data and field conditions:

Non-uniform backgrounds

PlantVillage images feature clean, plain backgrounds, whereas real-world images encompass soil, hands, shadows, or other plants. This discrepancy confuses CNN feature extraction.

Lighting variations

Field images may be influenced by direct sunlight, shadows, or low-light conditions, resulting in altered color distribution and rendering patterns more challenging to discern.

Leaf shape and orientation differences

Leaves in natural settings are rarely perfectly flat or centered, unlike dataset images, which affects feature alignment.

Dataset bias

The model was trained on controlled laboratory images, which diminishes its generalization to noisy environments.

These observations underscore the necessity for a more diverse training dataset and advanced augmentation strategies to enhance real-world accuracy.

5.7 Model Strengths and Limitations

Strengths

- **Lightweight MobileNetV2 architecture:** Enables fast inference and low memory usage.
- **Real-time analysis:** The Streamlit interface can process images immediately, making it suitable for field deployment.
- **User-friendly UI:** The application is accessible to non-technical users and provides actionable information.
- **Good generalization after fine-tuning:** The two-phase training approach significantly improved performance on challenging classes.
- **Actionable recommendations:** Disease-specific treatment suggestions assist in agricultural decision-making.

Limitations

- **Background Sensitivity:** Real-world backgrounds can significantly impact prediction confidence and accuracy.
- **Low Confidence for Noisy Images:** Certain field images may exhibit lower certainty due to variations in the dataset.
- **PlantVillage Dataset Bias:** The model's performance is optimized for uniform, well-lit images that closely resemble those present in the dataset.
- **Lack of Severity Estimation:** The system categorizes disease types but does not provide quantitative severity assessments.
- **No Multi-Leaf Handling:** The model assumes the presence of a single leaf within the frame.

6. CONCLUSION AND FUTURE WORK

6.1 Summary of Findings

This project aimed to design an intelligent and accessible system capable of classifying plant diseases using deep learning techniques. The primary objective was to automate early disease detection from leaf images, thereby supporting farmers, researchers, and agricultural practitioners in making informed and timely decisions. The work successfully demonstrated the integration of a fine-tuned MobileNetV2 model with a user-friendly Streamlit web interface, enabling real-time prediction and disease guidance.

The model was trained on the PlantVillage dataset, comprising 38 classes of healthy and diseased leaf images. By adopting a two-phase training strategy—initial frozen feature extraction followed by selective fine-tuning—the system achieved improved accuracy, stability, and generalization. The preprocessing pipeline, including resizing, normalization, and data augmentation, further enhanced the model's robustness. Training graphs, evaluation metrics, confusion matrix analysis, and classification reports indicated that the final model was capable of distinguishing between various disease patterns with reasonable accuracy.

The Streamlit web application serves as a crucial component of this project, providing an intuitive and interactive interface where users can upload or capture leaf images through a webcam. The application preprocesses the input image, performs inference using the trained model, and displays the top predictions along with confidence values. Additionally, disease descriptions and treatment recommendations were integrated to offer actionable guidance rather than mere prediction. This transforms the system into a practical decision-support tool rather than a simple classifier.

Overall, the project illustrates how deep learning models, when combined with modern deployment tools, can provide scalable solutions to agricultural challenges. Although real-world accuracy may vary due to uncontrolled environments, the system establishes a strong foundation for future improvements and real-field integration.

6.2 Conclusion

The escalating demand for efficient, accurate, and accessible plant disease detection methodologies has rendered AI-based solutions an indispensable component of contemporary agriculture. The completed project substantiates that deep learning models, specifically

MobileNetV2, can effectively classify plant diseases when trained on well-structured datasets. The Streamlit-based deployment further ensures that the solution transcends laboratory use, enabling accessibility to individuals with basic computer or smartphone camera capabilities.

The methodology employed in this project underscores several strengths. Transfer learning significantly mitigates the requirement for extensive labeled datasets while maintaining high performance. Fine-tuning selected layers enhances feature extraction and facilitates the model's adaptation to subtle variations in disease patterns. The user interface design prioritizes simplicity, facilitating effective interpretation and utilization of the system by individuals with no technical background.

Nevertheless, the project also highlights the challenges associated with domain shift—differences between controlled dataset images and real-world photographs. Factors such as lighting variations, background noise, partial leaves, and camera quality may impact prediction accuracy. Despite these limitations, the system presents a promising avenue for integrating AI into agricultural workflows and extending assistance to a broader user base.

In summation, this project provides a functional prototype for real-time plant disease detection utilizing deep learning and web technologies. It contributes to the burgeoning body of research in agricultural AI and establishes a foundation for the development of more advanced, scalable, and field-ready tools. With further enhancements, the system can evolve into a comprehensive agricultural diagnostic platform capable of supporting farmers in real-world challenges.

6.3 Future Enhancements

Despite the promising outcomes, there is substantial potential for enhancing the system's capabilities. Future research can concentrate on mitigating existing constraints and broadening the model's applicability to diverse environments and crop conditions.

6.3.1 Deploying on Cloud

Cloud deployment represents one of the most promising directions for improving scalability and accessibility. By hosting the model on cloud platforms such as AWS, Google Cloud, or Azure:

- Users can access the system from any device with an internet connection.
- Inference time can be reduced using GPU or TPU instances.
- Large-scale monitoring systems can integrate the model through APIs.

- Multiple users can access the system simultaneously without performance degradation.
- An automatic model update mechanism can be incorporated without requiring end-users to download new versions.

Cloud deployment facilitates the storage of user-submitted images, enabling the construction of future datasets that accurately represent real-world field conditions. This approach can substantially enhance model accuracy during retraining.

6.3.2 Adding More Plant Species

The current model is restricted to the 38 classes present in the PlantVillage dataset. To enhance its practical utility, the following expansions can be implemented:

- Incorporation of local or regional crop species not currently represented in the dataset
- Expansion of the disease types included, particularly those pertinent to specific geographic regions
 - Integration of real-field agro-dataset images
 - Support for multi-leaf or whole-plant images
 - Classification of disease severity (mild, moderate, severe)

Enhancing dataset diversity will not only expand the system's applicability but also assist in mitigating domain shift challenges that impact accuracy in uncontrolled environments.

6.3.3 Mobile Application Version

- Developing a mobile version of the system presents a significant future enhancement, considering the widespread adoption of smartphones by farmers and agricultural workers. A mobile application utilizing TensorFlow Lite can offer several advantages:
- Offline functionality without internet connectivity
- Enhanced inference speed on the device
- Optimized camera interface for leaf photography
- Immediate prediction delivery even in remote areas
- Integration of alerts, recommendations, and disease history tracking

A mobile application could also utilize GPS data to analyze disease trends regionally and provide community-based agricultural support services.

Additional Future Possibilities

Several other improvements can be explored:

- Explainable AI (XAI): Adding Grad-CAM visualizations to highlight infected regions on the leaf.
- Drone-based monitoring: Integrating the model with UAV systems for large-farm scanning.
- Farmer advisory platform: Combining predictions with fertilizer and pesticide suggestions.
- IoT integration: Leveraging sensors for multi-modal disease detection using environmental data.
- Active learning loops: Allowing the system to improve by learning from user-submitted images.

7. LIMITATIONS

Despite demonstrating promising results, the proposed plant disease classification system encounters several limitations that impede its performance in practical scenarios. The most significant limitations are enumerated below.

1. Dataset Bias (Controlled Backgrounds)

The PlantVillage dataset comprises images captured under controlled lighting and uniform backgrounds.

In contrast, real-world farm images frequently exhibit:

- Soil
- Shadows
- Cluttered backgrounds
- Varied lighting
- Partial leaves

This disparity results in diminished confidence and occasional misclassifications during real-world testing.

2. Sensitivity to Lighting and Shadows

The model's performance is most optimal when presented with clear, uniformly illuminated leaf images.

Conversely, strong shadows, overexposure, and low lighting conditions adversely impact the model's ability to:

- Accurately identify color features
- Recognize the texture of diseases
- Achieve precise segmentation

These factors contribute to inconsistent predictions.

3. Background Interference

The model was trained on centered leaf images without background noise. User-uploaded images often include:

- tables
- hands
- objects
- multiple leaves

This extra context confuses the convolutional filters and leads to reduced accuracy.

4. Limited to Visible Leaf Diseases

Only diseases with visible symptoms on leaf surfaces can be detected. The model cannot detect:

- soil-borne diseases
- stem/trunk infections
- nutrient deficiency without leaf symptoms
- early-stage invisible infections

5. Difficulty Distinguishing Similar Diseases

Some plant diseases have similar visual patterns, such as:

- Tomato Early Blight vs. Late Blight
- Leaf Mold vs. Septoria
- Yellow Leaf Curl vs. Mosaic Virus

These classes show overlap in color and texture, making them difficult for even experts to distinguish.

6. No Severity Estimation

The system predicts *what disease* is present but not:

- how severe the infection is

- whether treatment is urgent
- how far the disease has spread

Severity estimation would require segmentation or regression models.

7. Model Cannot Handle Multi-Leaf Images

If multiple leaves appear in the frame, the model assumes the dominant leaf is the target. It cannot classify:

- multiple plants
- overlapping leaves
- leaves with mixed symptoms

8. Limited Generalizability to All Plant Species

The model is trained on only 38 classes from the PlantVillage dataset. It cannot detect:

- diseases not present in the dataset
- rare crop species
- regional varieties not included during training

9. Requires Stable Internet or Local GPU for Deployment

Although Streamlit makes deployment easy, real-world farming scenarios may not have:

- reliable internet
- powerful devices for inference

Mobile deployment via TensorFlow Lite is not yet implemented.

10. No Explainable AI (XAI) Component

The system does not include:

- Grad-CAM heatmaps
- saliency maps
- explainability features

These tools help users understand why a prediction was made, increasing trust.

11. Confidence Scores May Be Low for Real Photos

Due to dataset background bias and variable lighting, confidence values sometimes remain below 10–20% even when the predicted class is correct.

This highlights the need for:

- more diverse training data
- improved augmentation
- domain adaptation

12. Model Performance Depends on Camera Quality

Low-resolution or noisy images from budget cameras can reduce performance significantly.

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